

De Novo Computational Design of Photoswitchable Cyclic Peptide Probes for F-actin

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Why image F-actin?

Filamentous actin (F-actin) is the principal structural component of the cytoskeleton and plays essential roles in cell migration, mechanotransduction, and neuronal development. Visualizing its dynamic organization in living cells is critical for understanding numerous biological processes.

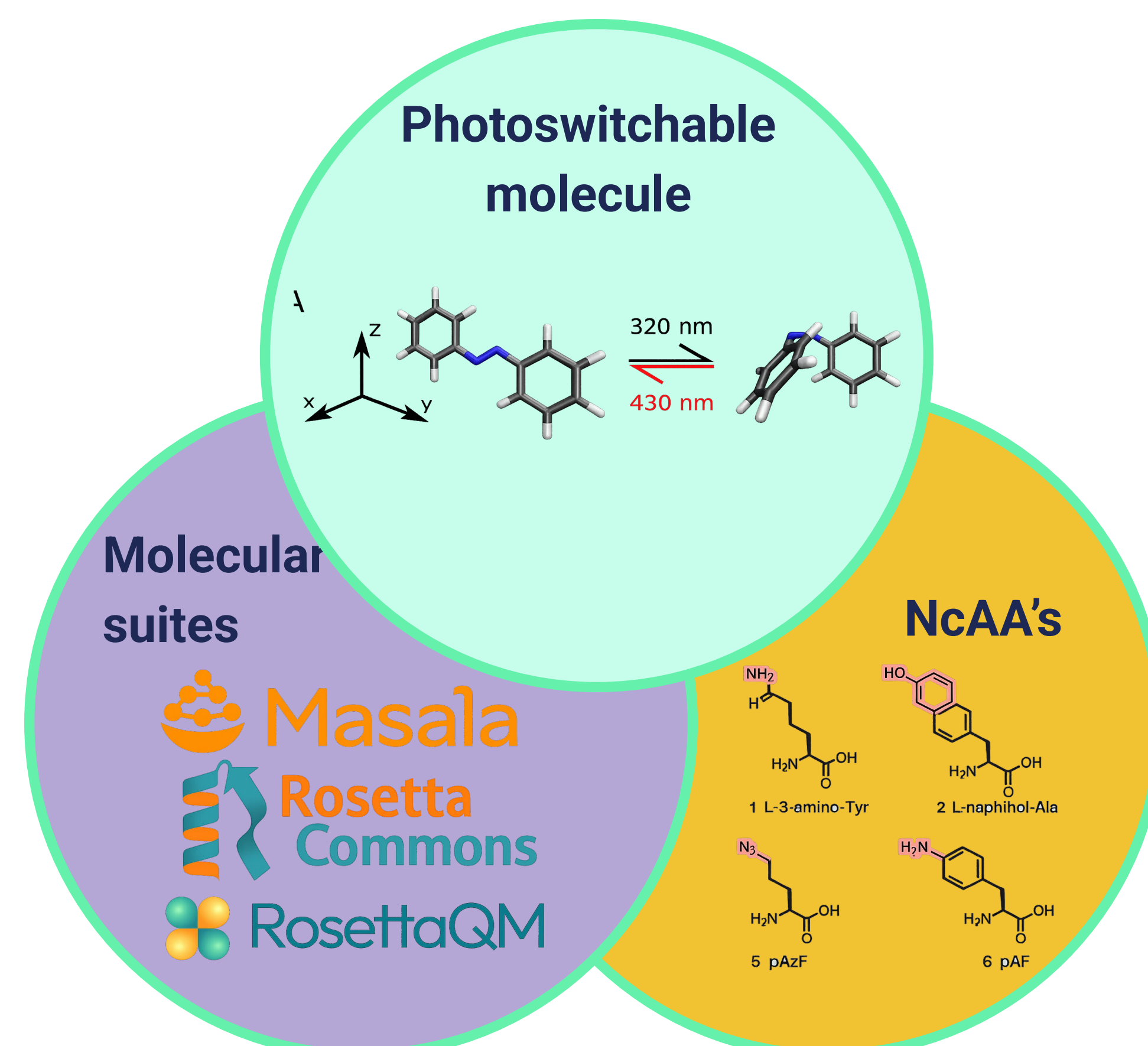


F-actin (PDB 8F8P)

Current limitations of F-actin probes

- Antibody-based labeling introduces spatial displacement.
- Limited control
- Competes with endogenous binders

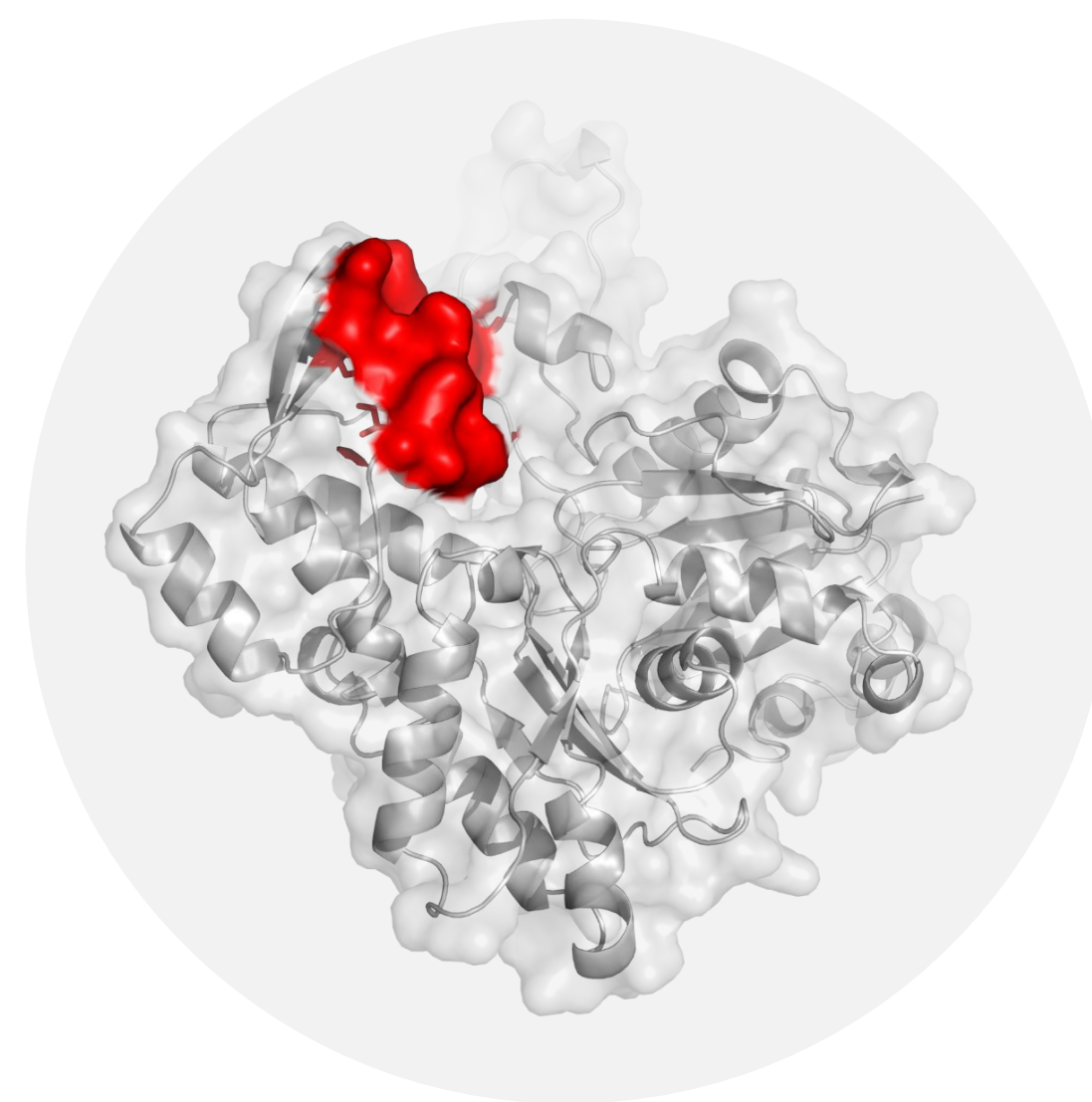
Photoswitchable Probes Components



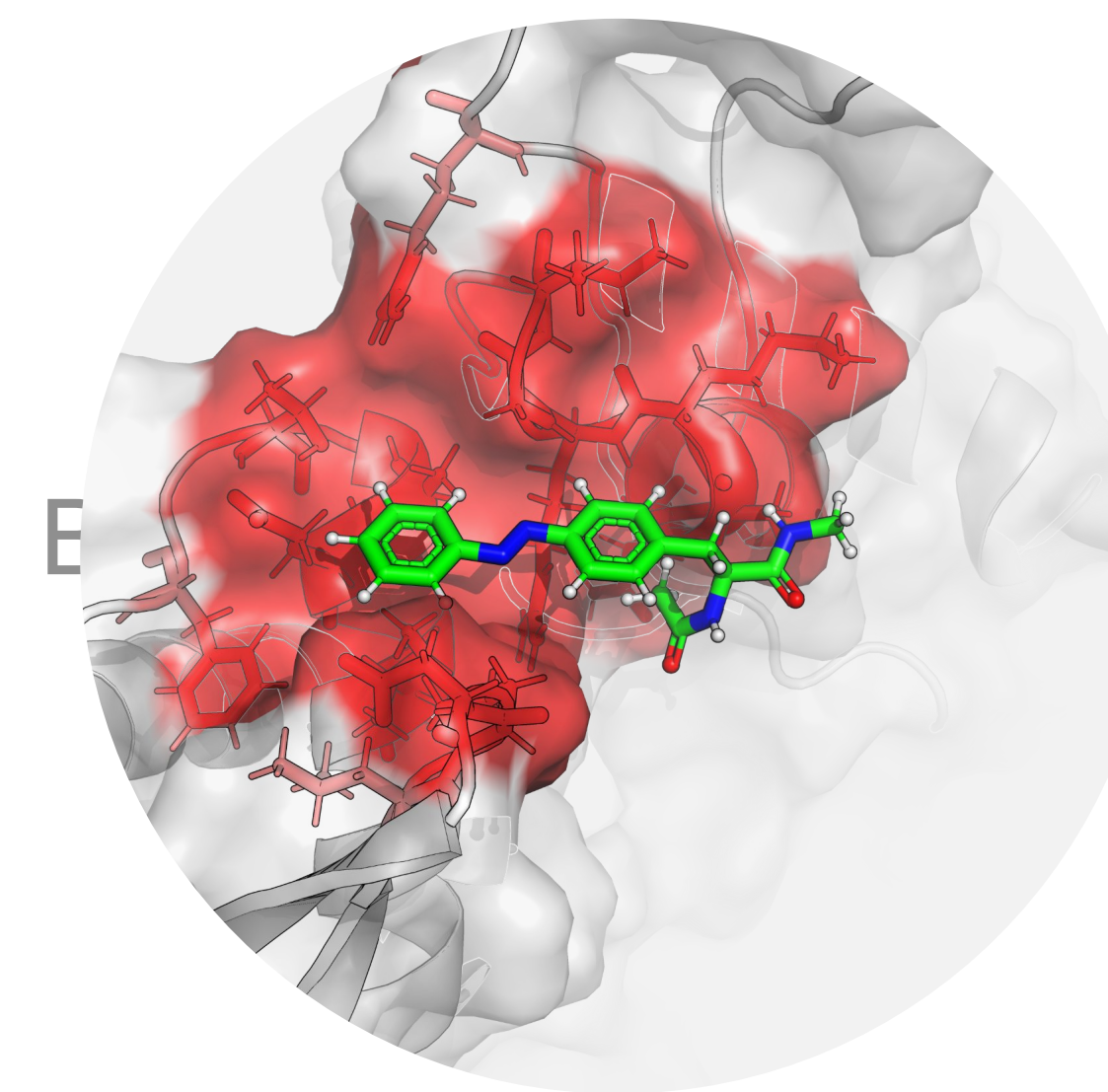
Azobenzene undergoes reversible cis–trans photoisomerization, enabling light-controlled changes in probe conformation.

Computational Design Strategy

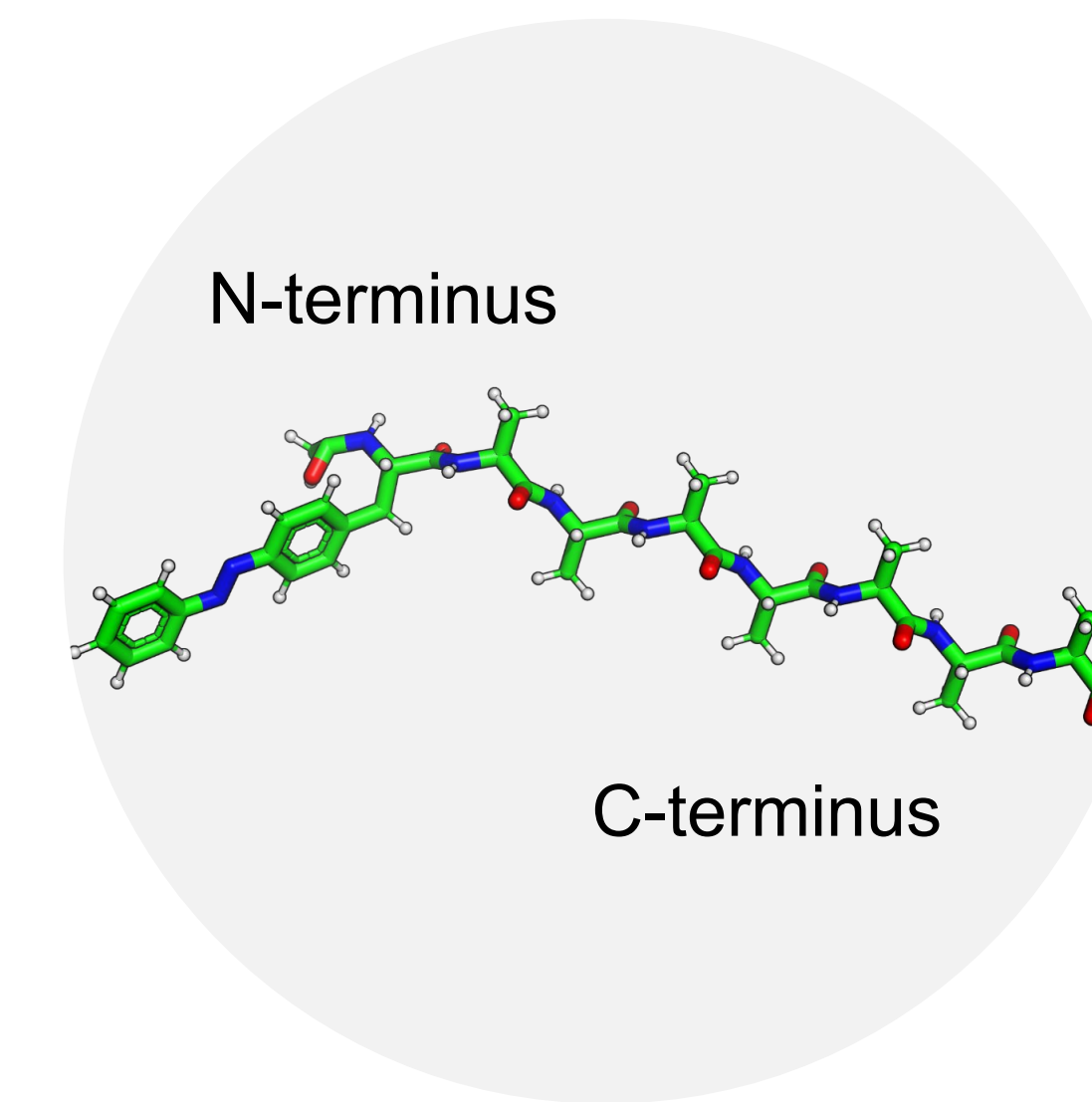
1. Site selection



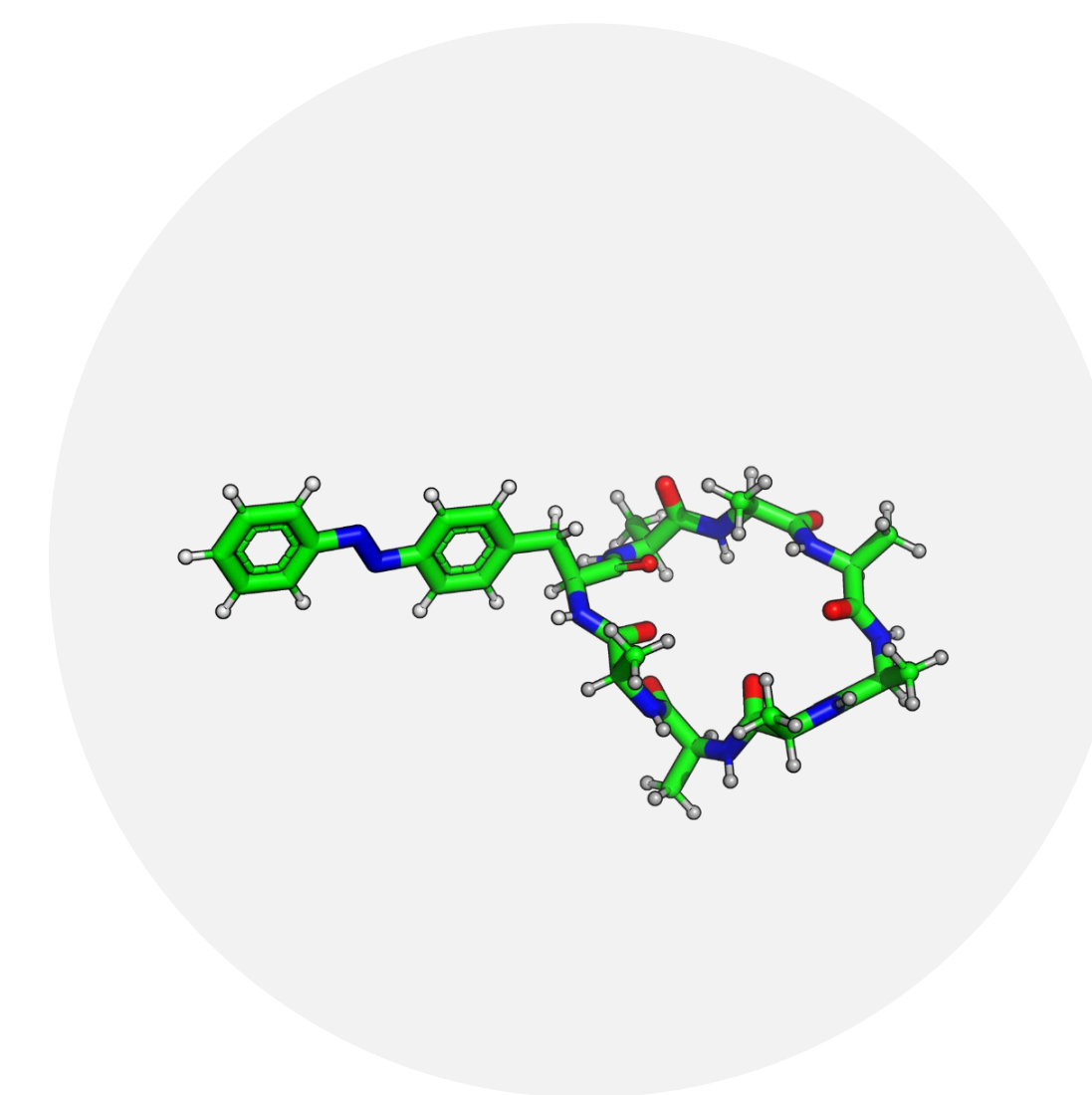
2. Photoswitch dock



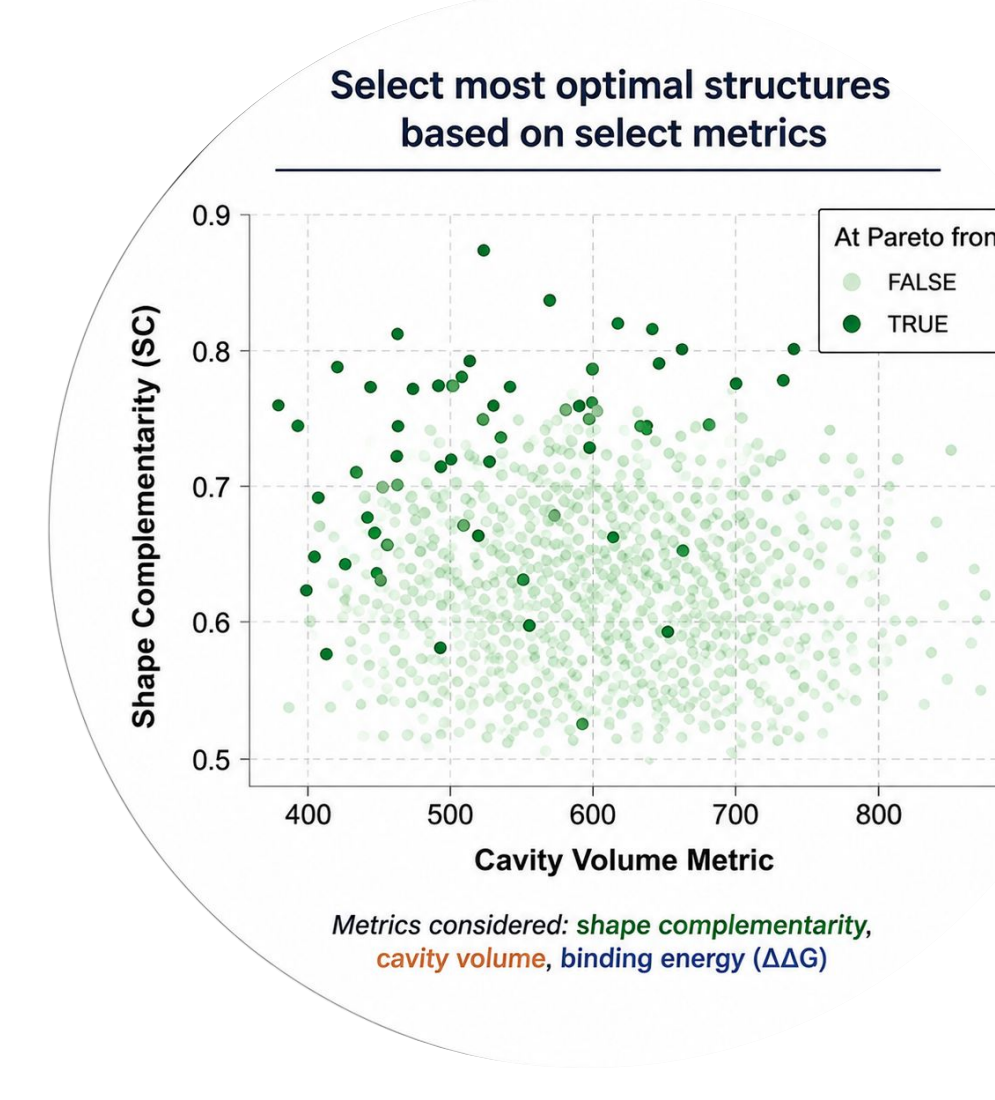
3. Probe extension



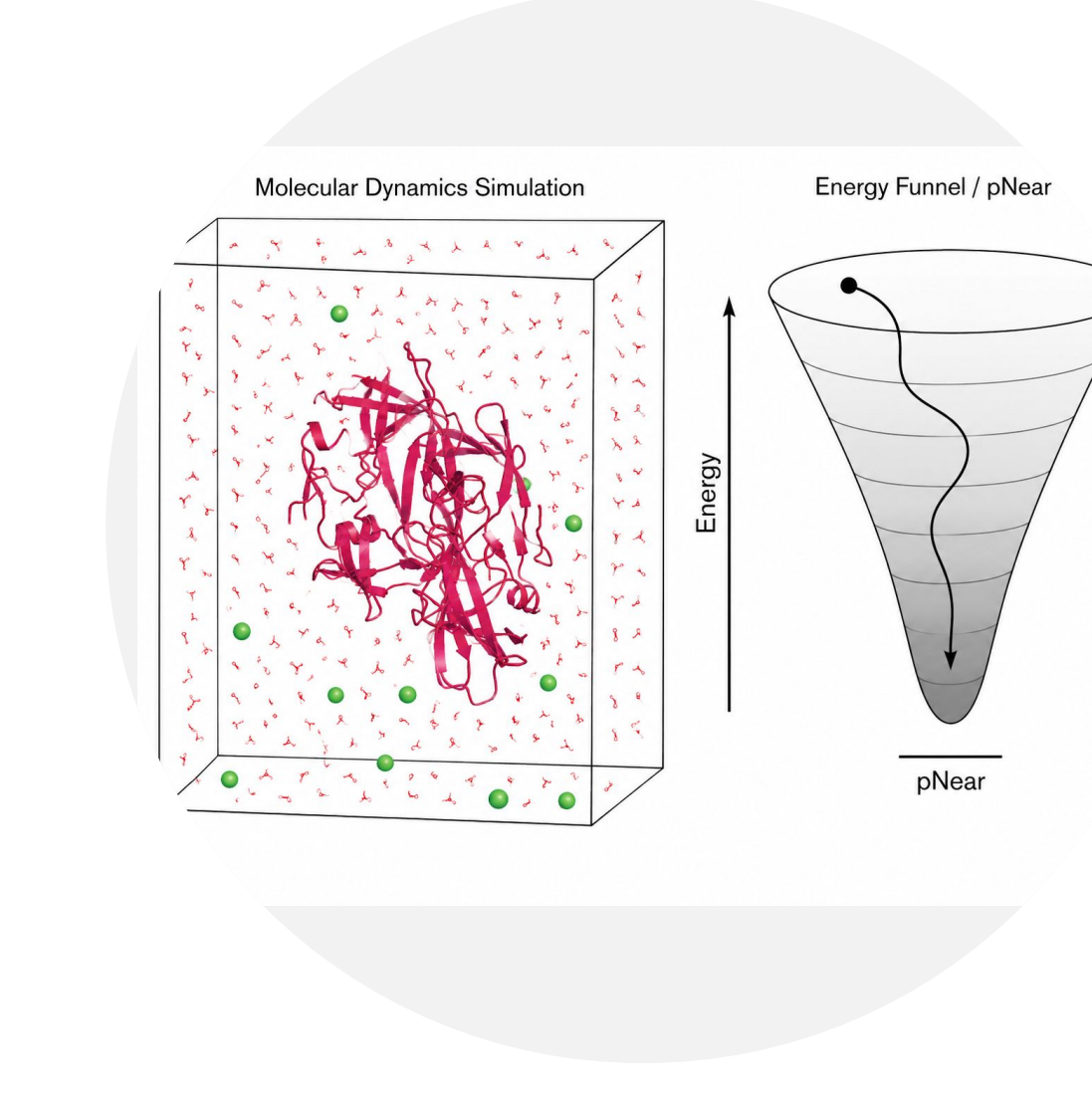
4. Cyclization & select lowest energy state



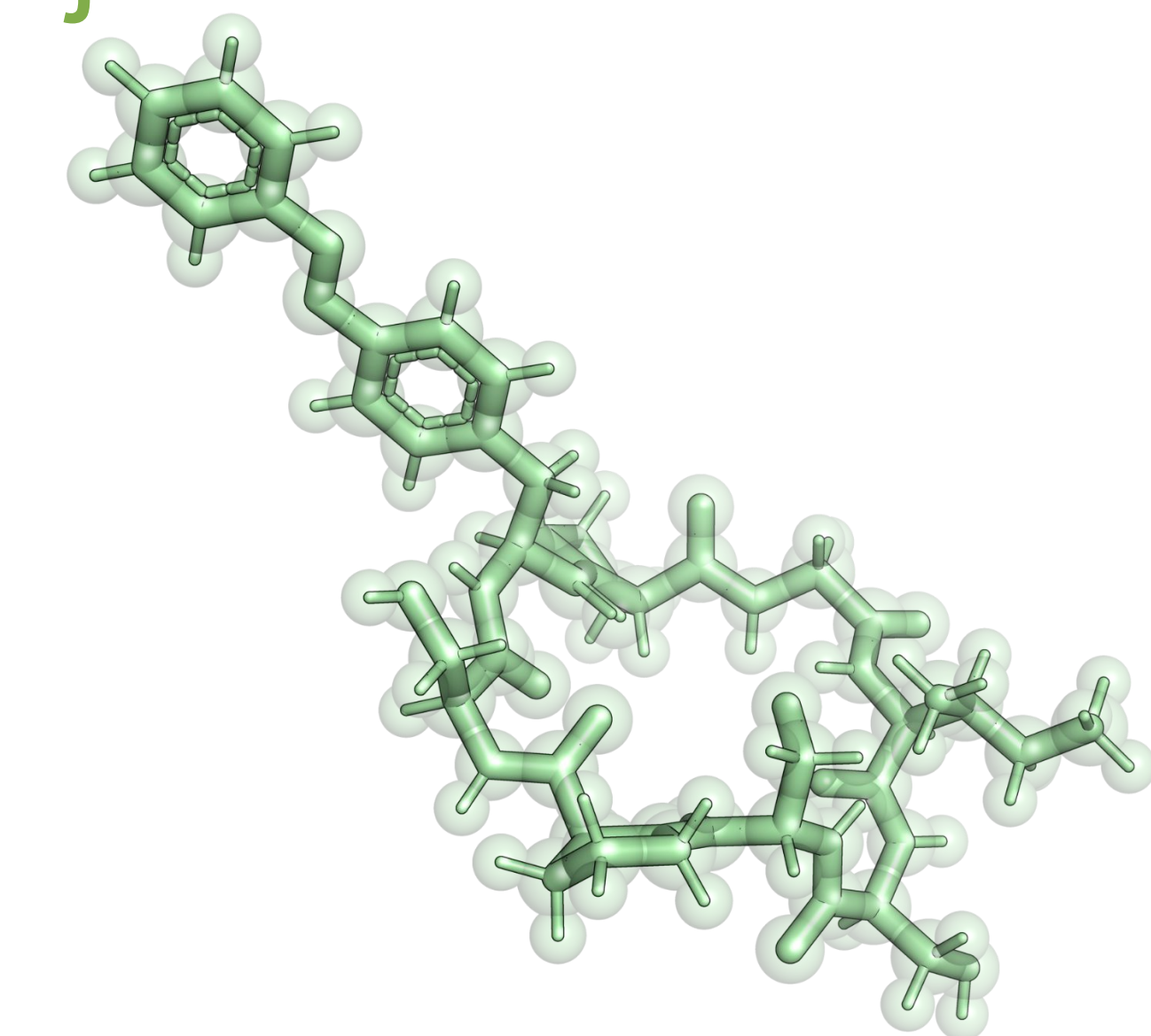
5. Select structure based on metrics



6. Evaluate Conformational preference & dynamics stability



Lead probe candidates (subject to final validation)



Trans conformation

Next steps in design

- pNear conformational validation
- Explicit-solvent molecular dynamics
- Trans–cis photoisomerization compatibility screening
- Lead selection for synthesis and experimental testing

Acknowledgements & References

Beharry AA, Woolley GA. Azobenzene photoswitches for biomolecules. *Chemical Society Reviews*. 2011;40(8):4422–4437.

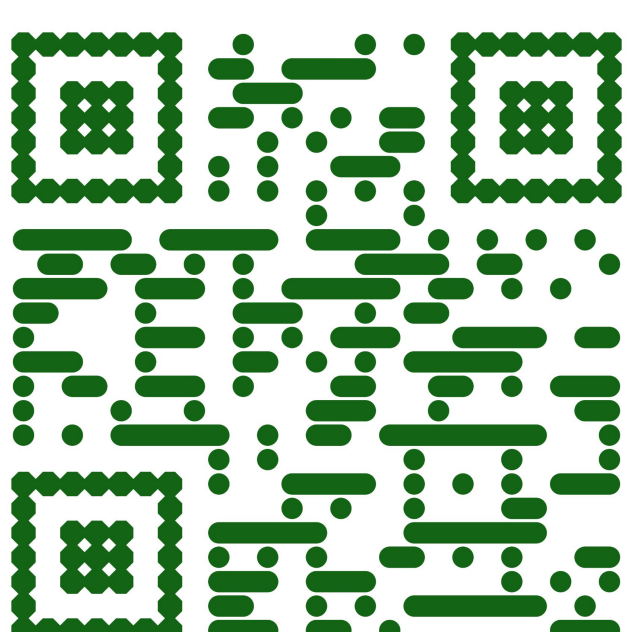
Leaver-Fay A, Tyka M, Lewis SM, et al. ROSETTA3: An object-oriented software suite for the simulation and design of macromolecules. *Methods in Enzymology*. 2011;487:545–574.

Zaborniak T, Azadvari N, Zhu Q, Turzo SMBA, Hosseinzadeh P, Renfrew PD, Mulligan VK. The open-source Masala software suite: Facilitating rapid methods development for synthetic heteropolymer design. *Methods in Enzymology*. 2025;723:299–426.

Computing Report:

The computational design and analysis were performed on the Flatiron Institute HPC infrastructure.

Scan to access the project repository:



In silico Validation of the probe

pNear analysis was used to evaluate whether low-energy conformations converged toward the designed cyclic structure. Higher pNear values indicate a more pronounced energy funnel and stronger conformational preference.

$$P_{\text{near}} = \frac{\sum_{i=1}^N \exp\left(-\frac{\text{r.m.s.d.}_i^2}{\lambda^2}\right) \exp\left(-\frac{E_i}{k_B T}\right)}{\sum_{j=1}^N \exp\left(-\frac{E_j}{k_B T}\right)}$$

